

KITILA, Finland

WMO: 027200

Lat: 67.696N

Lon: 24.858E

Elev: 196

StdP: 98.99

Time Zone: 2.00 (EUE)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-28.8	-35.2	0.1	-31.7	-31.8	0.2	-28.8	10.2	-4.1	9.2	-5.0	0.9	250	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	25.0	16.4	22.8	15.2	20.7	14.4	17.6	22.6	16.3	20.8	15.2	19.1	3.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.5	19.2	14.5	10.5	17.6	13.5	9.9	16.6	50.3	22.6	46.5	20.6	43.2	19.2	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.6	7.5	6.5	-35.9	27.0	2.8	2.0	-37.9	28.4	-39.6	29.6	-41.1	30.8	-43.2	32.3
			-36.0	19.0	2.8	1.5	-38.0	20.1	-39.6	21.0	-41.2	21.9	-43.3	23.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.3	-12.5	-12.3	-7.2	-0.4	5.9	11.0	14.6	12.1	7.4	0.0	-6.3	-8.9
	DBStd	10.77	7.56	8.02	5.97	3.96	4.46	3.75	3.50	3.34	3.49	5.03	6.13	6.86
	HDD10.0	3841	696	623	534	311	143	32	4	16	93	312	491	586
	HDD18.3	6577	954	857	793	561	386	221	125	196	329	570	741	844
	CDD10.0	317	0	0	0	0	16	62	145	81	13	0	0	0
	CDD18.3	11	0	0	0	0	0	2	8	2	0	0	0	0
	CDH23.3	134	0	0	0	0	6	20	91	17	0	0	0	0
	CDH26.7	16	0	0	0	0	0	2	11	2	0	0	0	0
(14) Wind	WSAvg	2.7	2.4	2.5	3.0	3.0	3.1	3.1	2.8	2.6	2.7	2.7	2.4	2.6
(15) Precipitation	PrecAvg	592	40	34	29	33	39	63	85	71	55	49	52	43
	PrecMax	758	78	76	61	75	96	106	141	144	112	93	96	76
	PrecMin	443	9	13	9	10	8	9	32	33	14	13	14	9
	PrecStd	92	15	16	14	16	20	23	30	28	27	21	21	18
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	4.1	6.9	13.9	24.0	26.2	28.2	25.8	17.9	11.1	6.0	4.2
		MCWB	0.5	2.2	2.7	7.4	13.8	17.4	17.6	17.7	12.6	9.2	5.8	2.5
	2%	DB	0.0	1.1	4.0	9.8	20.1	22.8	26.0	22.9	16.0	9.2	3.2	1.9
		MCWB	-0.4	-0.1	1.2	4.9	12.4	14.4	16.6	15.8	12.1	7.8	2.7	1.3
	5%	DB	-1.2	-0.2	2.1	7.8	16.7	20.0	23.9	20.6	14.3	8.0	1.8	0.8
		MCWB	-1.5	-0.7	0.1	3.6	10.1	13.1	16.0	15.0	11.2	7.1	1.5	0.3
	10%	DB	-2.8	-1.9	0.8	5.6	13.8	17.9	21.6	18.2	12.9	6.2	0.8	-0.2
		MCWB	-3.1	-2.3	-0.7	2.3	8.3	11.8	15.1	14.1	10.7	5.5	0.5	-0.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	2.4	2.9	7.5	14.3	18.3	19.5	19.0	14.2	9.7	5.6	2.7
		MCDB	1.0	4.1	5.9	13.1	21.9	24.5	24.7	24.0	16.3	10.4	5.6	3.9
	2%	WB	-0.4	0.2	1.5	5.0	12.6	15.8	17.9	17.1	12.9	8.0	2.7	1.3
		MCDB	-0.1	0.9	3.4	8.8	19.3	20.0	23.3	21.1	14.8	8.9	3.1	1.7
	5%	WB	-1.8	-0.8	0.4	3.7	10.5	14.0	16.8	15.6	11.8	6.8	1.2	0.2
		MCDB	-1.5	-0.3	1.7	7.1	15.7	18.2	21.8	19.0	13.7	7.7	1.4	0.4
	10%	WB	-3.1	-2.3	-0.5	2.6	8.8	12.7	15.8	14.6	11.0	5.6	0.3	-0.7
		MCDB	-2.8	-1.9	0.6	5.1	13.0	16.7	20.2	17.4	12.6	6.2	0.4	-0.4
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.1	9.7	8.3	8.6	8.2	8.9	9.2	7.3	5.6	6.0	7.5
		MCDBR	7.3	6.1	8.6	10.7	14.0	12.1	13.0	13.1	9.7	6.1	5.0	5.5
		MCWBR	7.1	5.5	6.5	6.7	7.2	5.7	5.9	6.9	6.2	4.9	4.8	5.2
	5% WB	MCDBR	7.3	6.1	7.3	9.8	12.6	9.8	10.9	11.1	7.5	5.5	4.7	5.5
		MCWBR	7.2	5.5	5.8	6.4	7.0	5.5	5.6	6.4	5.2	4.6	4.6	5.3
(40) Clear-Sky Solar Irradiance	taub		0.189	0.251	0.286	0.317	0.335	0.338	0.348	0.333	0.304	0.271	0.349	N/A
	taud		1.925	2.094	2.182	2.216	2.429	2.485	2.468	2.493	2.550	2.336	2.740	N/A
	Ebn at Noon		129	555	751	830	862	868	843	819	760	550	171	0
	Edh at Noon		10	49	80	103	94	91	90	79	58	39	13	0
(44) All-Sky Solar Radiation	RadAvg		0.04	0.52	2.02	3.91	4.82	5.01	4.67	3.42	1.79	0.60	0.10	0.00
	RadStd		0.00	0.07	0.15	0.23	0.47	0.37	0.46	0.28	0.19	0.11	0.01	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page